

CUDA PARALLEL ARCHITECTURE AND PROGRAMMING MODEL

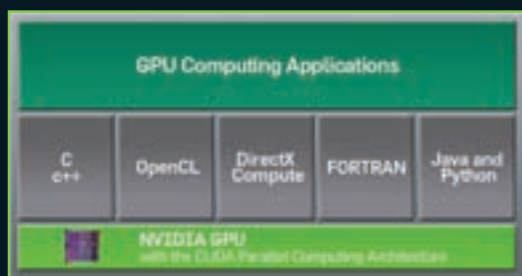
The problem was that GPGPU required using graphics programming languages like OpenGL and Cg to program the GPU. Developers had to make their scientific applications look like graphics applications and map them into problems that drew triangles and polygons. This limited the accessibility of tremendous performance of GPUs for science.

NVIDIA realized the potential to bring this performance to the larger scientific community and decided to invest in modifying the GPU to make it fully programmable for scientific applications and added support for high-level languages like C, C++, and Fortran. This led to the CUDA architecture for the GPU.

The CUDA parallel hardware architecture is accompanied by the CUDA parallel programming model that provides a set of abstractions that enable expressing fine-grained and coarse-grain data and task parallelism. The programmer can choose to express the parallelism in high-level languages such as C, C++, Fortran or driver APIs such as OpenCL™ and DirectX™-11 Compute.

The CUDA parallel programming model guides programmers to partition the problem into coarse sub-problems that can be solved independently in parallel. Fine grain parallelism in the sub-problems is then expressed such that each sub-problem can be solved cooperatively in parallel.

The CUDA GPU architecture and the corresponding CUDA parallel computing model are now widely deployed with 1000s of applications and 1000s of published research papers. CUDA Zone lists many of these applications and papers.



REAL WORLD APPLICATIONS

With millions of CUDA-enabled GPUs sold to date, software developers, scientists and researchers are finding broad-ranging uses for GPU computing with CUDA. Here are a few examples:

Identify hidden plaque in arteries: Heart attacks are the leading cause of death worldwide. Harvard Engineering, Harvard Medical School and Brigham & Women's Hospital have teamed up to use GPUs to simulate blood flow and identify hidden arterial plaque without invasive imaging techniques or exploratory surgery.

Analyze air traffic flow: The National Airspace System manages the nationwide coordination of air



QUADRO Q6000 IS BUILT WITH ADVANCED FEATURE SETS INCLUDING AROUND 6 GB OF FRAME BUFFER, CAN PROCESS AROUND 1.3 BILLION TRIANGLES PER SECOND

traffic flow. Computer models help identify new ways to alleviate congestion and keep airplane traffic moving efficiently. Using the computational power of GPUs, a team at NASA obtained a large performance gain, reducing analysis time from ten minutes to three seconds.

Visualize molecules: A molecular simulation called NAMD (nanoscale molecular dynamics) gets a large performance boost with GPUs. The speed-up is a result of the parallel architecture of GPUs, which enables NAMD developers to port compute-intensive portions of the application to the GPU using the CUDA Toolkit.

THE RIGHT CARDS

You may be wondering that with such a change in the use case scenarios - from pure gaming to visualization and computing - which cards would be the right fit.

Quadro series of cards from NVIDIA fit that bill. If you are the kinds who uses a lot of 3DSMax or Maya, then you are bound to need a lot of graphics horsepower for faster rendering of your 2D images. Quadro cards, for instance the Quadro Q6000 are built with advanced feature sets including around 6 GB of frame buffer, can process around 1.3 billion triangles per second, etc. These are capabilities that work well for professional applications for instance larger onboard memory means that you can render larger and more complex scenes. Also with a processing power of over a billion triangles per second, the rendering will be much faster on a Quadro card than a GeForce one, and definitely at much lower load temperatures than GeForce cards - thereby increasing the life cycles of such cards. A Quadro card comes with a 3 year warranty as opposed to a standard GeForce gaming card which has a 1 year warranty.

Pricing of these professional cards is much higher than your standard gaming cards. Geforce cards are overclocked so as to operate on higher frequencies this increases the performance but reduces the reliability and life of the card whereas Quadro cards are clocked at such a frequency so as to achieve good balance between performance and reliability. 